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Youth AI designs meet eternal values: responsible digital technology design and innovation education

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ABSTRACT

Critical and reflective thinking towards technology should be fostered in adolescents, who represent future citizens with influential professions. Prior research has proposed an understanding of business as a crucial aspect of this endeavour. Our study explores how to cultivate responsible business thinking among adolescents in relation to the design and use of future technologies. To achieve this objective, we analysed data generated by 12 adolescents aged 15–17, as they created business ideas for sustainable innovations leveraging artificial intelligence. We examined their perceptions of technology and business within a value framework inspired by Zoroastrianism, a religion with a value base that has influenced cultures, other religions, and civilisations globally. Drawing from our findings, we propose key elements for responsible digital technology design and innovation education for adolescents, holding the potential to enhance their critical thinking and ethical engagement with future technologies.

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Adolescents; computational empowerment; innovation education; design/technology protagonist; Zoroastrian values

1 Introduction

Technology plays a central role in our lives. Children, future adults, are active technology users, and recent studies call for their computational empowerment, i.e. their ‘active engagement in matters related to digital technology and its effects on society’ (Iivari et al. 2024). Linked with this, the concept of a ‘design protagonist’ (Iversen, Smith, and Dindler 2017), has emerged, entailing the ability to consider ethical and reflective design and use of technology. In addition to design skills, innovation skills have also been linked with being a design protagonist (Kinnula, Durall, and Haukipuro 2022). After all, technology design and business go hand in hand: companies create innovations that, in turn, drive new business opportunities. Given the overlap of technology and business, it is crucial to consider the implications of the technologies and business models we develop. Moreover, it is crucial to focus on ‘solving big issues’ for the current and future generations (e.g. Bødker and Kyng 2018). This implies a need for a critical and reflective mindset towards both technology and business (Hartikainen et al. 2023). Indeed, more research is needed into how to educate business-oriented design protagonists with responsible mindsets (Hartikainen et al. 2023).

It has been argued that we co-evolve with technology – on the one hand, humans make changes in the world by creating new technologies, but on the other hand, technologies also change us (Haddington et al. 2021). With AI increasingly influencing society, it is essential to educate adolescents about its ethical and societal implications (Durall Gazulla et al. 2024). There is already a lot of work focusing on ethical artificial intelligence (AI) education through school curricula, projects, or other methods, such as game-based learning (Ali, Kumar, and Breazeal 2023; Park and Kwon 2024) to teach the societal impacts of AI systems. Entrepreneurship research has long been interested in ethics and values as well (Hannafey 2003; Harris, Sapienza, and Bowie 2009; Vallaster et al. 2019), and this kind of work is especially needed when focusing on AI and the development of AI systems (Hagendorff 2020; Jobin, Ienca, and Vayena 2019). After all, technology is never value-neutral (Martin and Freeman 2004) and AI will be an integral part of our future.

Existing studies on ethical AI education primarily focus on curriculum design and formal projects; however, few address the intersection of adolescent values and AI innovation. In the current study, our aim is to explore how to cultivate responsible thinking among

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adolescents in relation to the design and use of emergent technologies. We study this in the context of sustainable innovations that utilise AI. We organised a project of design and innovation education in the form of a two-week AI Ambassador programme at a local university, aiming to teach AI literacy to teenagers through hands-on activities on AI-related topics. As a part of the programme, 12 adolescents aged 15–17 created sustainable business ideas leveraging AI. As our research question we ask:

How the Zoroastrian-inspired values lens can inform strategies to foster responsible digital technology innovation in future design protagonists?

We believe that understanding adolescents' values in technology-related innovation is essential for fostering responsible future designers. For our analysis, we take inspiration from Zoroastrianism. As one of the oldest religions and social philosophies, Zoroastrianism has influenced Christianity, Islamic thinking, and Buddhist philosophy (Stausberg 2008). By using a Zoroastrian-inspired ethical lens, we are not only drawing from one particular religion but also on eternal values that live through other philosophies alive today. We think the Zoroastrian value system resonates with the current strong focus on sustainability questions and provides an interesting basis to examine responsible innovations.

2. Related research

2.1. Children as innovators of digital futures

As children's relationships with technology evolve, researchers advocate for technology design by children and child-driven design to ensure children are empowered to make a change in their lives and the surrounding technology-rich world (Iversen, Smith, and Dindler 2017; Kinnula and Iivari 2021). The aim is to establish children as experts in their lived experiences and protagonists of their technology encounters. This is argued to empower them to shape and influence technologies (Iversen, Smith, and Dindler 2017; Kinnula and Iivari 2021), even drive the process where designing is done by them (and not with or for them).

From the perspective of educating these design protagonists to act as innovators of digital futures, integration of innovation education with technology education is considered beneficial in teaching many twenty-first-century skills (Wilson et al. 2009; World Economic Forum n.d.). So far, innovation education has predominantly been explored in the context of higher education studies, and there is a scarcity of research on the intersection of children and adolescents' technology education and innovation education. Existing studies have addressed

various facets of this field, such as finding out if students learn innovation in comprehensive education (Lindfors and Hilmola 2016), youths' sustainability thinking in their innovations (Kinnula, Durall, and Haukipuro 2022), design ideas from the product concept (business and innovation) perspective (Read et al. 2016), challenges faced by adult educators (Hartikainen et al. 2021), struggles children might encounter in conceptualising future product users (Ventä-Olkkonen et al. 2019, 2021), the organisation and associated challenges of the learning process (Hartikainen et al. 2021; Hollauf et al. 2020), and the significance of introducing entrepreneurship as a career option (Geser et al. 2019; Paço and Palinhas 2011).

Hollauf et al. (2020) outline three dimensions for fostering social innovation projects with youth relating to social innovation, entrepreneurial learning, and digital technology design and making, and seek ways to motivate learners within these dimensions. Iivari, Molin-Juustila, and Kinnula (2016) suggest organisers should pay close attention to both intrinsic and extrinsic motivational factors that influence learners, in order to scaffold learners' participation. They advocate adopting a collective activity system perspective and establishing purposeful, shared project goals to align the interests of each participant toward achieving those objectives. The role of makerspaces, such as FabLabs, in design and innovation education has gained recognition, being seen as a catalyst for innovation and a fundamental element of entrepreneurship (Hartikainen et al. 2021; Hartikainen et al. 2023; Ventä-Olkkonen et al. 2021). From the perspective of innovation education, Kinnula, Durall, and Haukipuro (2022) highlight the need for children to gain systems thinking capabilities, noting that business ideating within the context of sustainable technology innovations appears to be suitable for developing those capabilities. Generally, fostering the development of skills valuable for aspiring innovators/entrepreneurs, including qualities like creativity, entrepreneurial spirit, and independence, has been argued to serve as a foundation that can be built upon in subsequent stages of education (Geser et al. 2019; Paço and Palinhas 2011).

In the current study, we approach fostering responsible digital technology innovations through the lens of values reflected in the business ideas created by teenagers. Focus is not on the business ideas as such, but the values they highlight. Next, we will discuss values related to design.

2.2. Values in design

Values play an important role in driving human actions, serving as benchmarks for evaluating and rationalising

our behaviour (Schwartz 1992, 1994). They tell us what is deemed important, virtuous, and morally correct in life (Friedman et al. 2013; Iversen, Halskov, and Leong 2010; Miller et al. 2007). Values can be seen as desirable goals that act as guiding principles for both individuals and communities (Schwartz 1992). The importance of values shaping and underlying technology design and use has already been widely acknowledged in general – technology is never value-neutral (Martin and Freeman 2004) – and it has also been recognised by researchers working with children that values related to technology should be addressed together with children (e.g. Hartikainen, Iivari, and Kinnula 2019; Iivari, Kinnula, and Kuure 2015; Isomursu et al. 2011; Iversen, Halskov, and Leong 2010; Iversen and Smith 2012; Kinnula et al. 2018; Nouwen, Van Mechelen, and Zaman 2015). Researchers have been engaging in, e.g. discussing how underlying values might affect design activities with youth (Iversen and Smith 2012), exploring values of stakeholders like industry representatives, parents and children related to parental control software (Nouwen, Van Mechelen, and Zaman 2015; Rutkowski et al. 2021; Wisniewski et al. 2017), mapping how children's values could be embedded, e.g. in online safety education targeting them (Hartikainen, Iivari, and Kinnula 2019) or location sharing apps parents use (Kayal et al. 2018). Researchers have also engaged children in identifying values that might be embedded in current technologies such as digital games (Kynigos and Yiannoutsou 2018), or values they might embed in their own technology designs (Hartikainen et al. 2023; Iivari et al. 2023).

Several approaches, models, and tools have been developed to explore and incorporate values into technology and design-related research. Key methodologies include e.g. the Values at Play Framework, the HuValue tool, Worth-Centered Design, Values-led Participatory Design, and Value-Sensitive Design. *The Values at Play framework* (Flanagan, Howe, and Nissenbaum 2008) is oriented around ethical, social, and political values. It promotes values integration through a three-step process: discovery (identifying relevant values), translation (embedding these values into the design), and verification (evaluating how well values have been implemented in the final product). *HuValue* is a tool that intends to raise designers' awareness about human values and provide tangible materials to facilitate the use of selected values within the design process (Kheirandish et al. 2020). *Values-led Participatory Design* (Leong and Iversen 2015) fosters the emergence and development of relevant values through active engagement with stakeholders. This method begins with a focus on understanding participants' values and nurtures them throughout the design process via

ongoing dialogue. *Worth-centered Design* (Cockton 2006) aims to create products that are inherently valuable to users. This approach includes analysing stakeholder needs, designing prototypes that reflect meaningful attributes, evaluating their perceived worth, and iterating based on feedback. Finally, *Value-Sensitive Design* (Friedman et al. 2013) integrates ethical considerations into technology development through three levels of investigation: conceptual (exploring foundational constructs), empirical (examining social contexts and user interactions), and technical (analysing systems to assess value alignment). Together, these methodologies emphasise the role of human values in shaping technology that is ethically aware, socially responsive, and genuinely beneficial to users.

Indeed, investigating values in digital technology design is essential because values guide human behaviour, influence perceptions of what is ethically and socially right, and serve as benchmarks for assessing decisions and action. In technology, values shape both the purpose and impact of design, underscoring the need for tools and frameworks that incorporate ethical, social, and cultural considerations. Behind our personal values are cultural, religious, and philosophical traditions. A large part of digital technology design-related research has been traditionally grounded on frameworks and practices derived from Western values. In recent years, the field has, however, become more polyphonic and inclusive, while different indigenous epistemologies have been taken into account (e.g. Winschiers-Theophilus and Bidwell 2013). In addition to indigenous epistemologies, different religions' ideological and values lenses are also gaining more emphasis. Inspired by this body of research, we next briefly review existing digital technology design related research that takes inspiration from religions, particularly Zoroastrianism, before moving on to present our theoretical lens in detail.

2.3. Religions as inspiration in digital technology design

Next to Christian values-based studies in the Western world (e.g. Ames, Rosner, and Erickson 2015; Laato, Rauti, and Hamari 2023; Wyche and Grinter 2009) studies based on especially Islamic philosophy have gained prominence (Ibtasam 2021; Rifat, Toriq, and Ahmed 2020, 2022) but also, for instance, Asian religions and philosophies are present in the studies (e.g. Huang and Lin 2021). Religion inspired studies have concentrated on *understanding users* from different religious cultures (Ibtasam 2021), and for *guiding designers* to take into account different religions, beliefs, and their practices

(Rifat, Toriq, and Ahmed 2020, 2022; Wyche and Grinter 2009), to harness these philosophies in designing better products (Huang and Lin 2021), and for studying how religion-based *beliefs affect designers and engineers* themselves (Ames, Rosner, and Erickson 2015). For example, Ibtasam (2021) focused on how religion forms users' individual and collective values and socio-cultural norms that impact users' understanding, use, or perception of technologies, and Rifat et al. (2022) ask how human-computer interaction (HCI) design can address the needs and values of faith-based communities and how scholarship and practice can benefit from the integration of faith, religion, and spirituality.

Whereas the biggest world religions are present in this body of research, so far, studies inspired by Zoroastrianism in the field of digital technology design have been scarce. For example, in November 2024, Zoroastrianism was mentioned only in 18 publications in the ACM Digital Library. In all these publications, the word was mentioned only once. The oldest mention was from 1984 (Conrad and Rahimi 1984), and the rest were from between 2012 and 2024. The term is mostly mentioned in background information listed with other religions e.g. while describing the study participants (Shrestha et al. 2024), the context (Khan and Zaber 2022; Neha et al. 2022), the origin of a central concept of the study (Naamneh et al. 2023). In Vaida (2019), it is mentioned while describing the background of a collaborative team development tool. None use Zoroastrianism and its values as inspiration for analysis or apply them. The only exception is (Nagda et al. 2024) where the system was named after Zoroastrian deity of justice. Even though Zoroastrian values have not guided digital technology design endeavours, examples can be found in other fields where the values lens has been utilised successfully. The positive impact of Zoroastrianism on tending of e.g. trees, forests and animals is represented (Yachkaschi and Yachkaschi 2012), and it has been used when addressing a water crisis (Foltz 2002).

While the above-mentioned approaches illustrate how religion can inform design thinking, the Zoroastrian worldview offers a distinctive contribution by linking ethical intention, truthful action, and environmental stewardship within a coherent moral order. In the following section, we use this worldview as an analytical lens to extend existing value frameworks beyond procedural and human-centred concerns.

3. Zoroastrian worldview as an analytical lens for the study

The Zoroastrian worldview, rooted in ancient values of integrity, harmony, and human responsibility, offers a

strong framework for analysing ethical considerations in technology and business innovation. Central to Zoroastrianism is the duality between Asha representing truth and order and Druj, representing deception and chaos (Boyce n.d.; Cantera 2015; Dhalla 1938; Nigosian 1993). Humanity's duty is to uphold Asha through choices that reflect 'good thoughts, good words, good deeds', a triad representing the values of integrity, cooperation, wisdom/sapience, balancing physical and spiritual welfare and environmental stewardship. The worldview fosters individual agency, societal harmony, and a collective responsibility to sustain both the social and natural world (Boyce n.d.; Cantera 2015; Kreyenbroek n.d.). This worldview not only tasks humans with the establishment and maintenance of a just society but also underscores their responsibility for environmental stewardship. In doing so, Zoroastrian value system highlights human agency, the transformative potential of righteous choices, and the broader societal and environmental implications of individual decisions.

1. Understanding good through the following values:

- **Integrity:** The extent to which people are consistent in their moral actions and uphold the truth and correctness in their lives; emphasises truth and honesty, guiding actions authentically. Our interpretation of this is that we should aim for truth and righteousness in actions.

- **Cooperation:** How we contribute towards collective good, take responsibility for one another, and build harmonious relationships with others; reflects societal harmony, focusing on collaborative efforts and support. We should aim for cooperation and societal harmony.

- **Sapience:** Reflects the aim of critical thinking, enlightened decision-making, and continual development and learning; prioritises wisdom's development for growth and understanding. We should aim for wise choices.

- **Wellbeing:** A holistic view of welfare, self-actualisation, and balance in life; balances physical and spiritual welfare. We should aim for the well-being of everybody.

- **Protection of the environment:** The stewardship of nature to the benefit of everyone and the interconnectedness of human existence with the natural world; stresses humanity's role as stewards for the environment. We should consider the impact on the environment in our actions.

2. Cosmic harmony and societal and environmental responsibility:

- This principle reflects Zoroastrianism's emphasis on humanity's role in preserving balance in the struggle between Asha and Druj – truth and order vs. deception



Figure 1. United Nations Sustainable Development Goals.

and chaos. We see this aligns with the need to solve ‘big issues’ facing society (see e.g. Bødker and Kyng 2018). We should ourselves responsible to uphold the cosmic harmony and make conscious decisions when doing everyday things.

– The United Nations (UN) has defined 17 Sustainable Development Goals (SDGs)¹ address global challenges and ensure a better future (Figure 1). Taken together, SDGs reflect the global commitment to a sustainable, inclusive, and better future. Therefore, we think they represent the contemporary and secular depiction of the Zoroastrian cosmic struggle toward a harmonious world. We used the SDGs as secular representations of Zoroastrian ideals to support this dimension of our analysis.

3. **Ethical triad** – holistic alignment of thought, speech, and action:

– The Zoroastrian ethical triad emphasises coherence between intention, communication, and conduct, resonating with values of authenticity and purpose in design. We should aim for good in thoughts, speech, and actions; to act with ‘good thought, good words, and good deeds’. Good thought reflects the purity of mind and intentions behind the actions. It revolves around emphasising positive and constructive ways of approaching the world. Good words cultivate speaking truthfully but also kindly and positively to others. Good deeds highlight the moral, kind, and constructive actions toward the betterment of the world

We think using Zoroastrian values in digital technology design could foster ethical decision-making, encourage social responsibility, and inspire sustainable innovations that align with its humanistic and environmental ideals. Therefore, integrating Zoroastrianism or other value-

based lenses can support a more inclusive, ethically grounded approach to technology.

4. Research design

4.1. Participant selection

The study was part of a two-week AI Ambassador programme at a local university, aiming to teach AI literacy to adolescents through hands-on design, innovation, and reflective activities on AI-related topics. Participants aged 15–17 were selected on a first-come, first-served basis through city summer worker calls. In the end, we had 12 participants (called AI ambassadors from now on) with different skills and backgrounds. Most were highly familiar with terms like image recognition, chatbots, and algorithms, while some had moderate familiarity but lacked comprehensive understanding, while some felt less confident. Some showed deeper awareness by mentioning specific AI technologies such as Dall-E, deep learning, and neural networks.

The AI ambassadors were divided into four teams of three participants, each based on their self-declared interests on their first workday. We did not assign them any roles in the teams; they negotiated their roles on their own. The teams and the participants are described in more detail in Table 1; their backgrounds have been combined, and business idea names have been changed to maintain anonymity. Note that we did not consider demographic information to be relevant as the participants were self-selected by the application process. Moreover, the sample was small and aimed towards qualitative analysis rather than generalisable results. Therefore, we did not ask for information

Table 1. AI ambassadors.

Team	Business idea	Participants' age and gender	Participants' backgrounds combined
1	Vitalis: An app for wellbeing with personalised strategies, linked to own medical data.	17, girl 17, girl 17, boy	Prior experience in programming, digital fabrication and making, customer service, community service, content creation, student council, game-making, mathematics competitions
2	SortAI: A waste management system that utilises AI for efficient sorting and recycling.	17, girl 16, boy 16, boy	Prior experience in programming, web design, digital fabrication and making, IT, customer service, warehouse work, volunteer work, music, competitive swimming, student council
3	Careo: A patient-monitoring healthcare robot to support healthcare professionals.	17, girl 15, girl 16, boy	Prior experience in digital fabrication and making, content creation, customer service, tutoring, athletics coaching, volunteering, music, gaming, working at the university
4	Adminix: An app to handle hospital administration and management to reduce burden on healthcare professionals	17, boy 16, boy 15, boy	Prior experience in data entry and analysis, programming, 3D graphics, digital fabrication and making, digital art, customer service, role-playing games, swimming

about gender. Instead, the gender categorisation is assumed.

This paper focuses on the two 6-hour workdays of the programme (altogether 12 h) during which the AI ambassadors ideated and planned sustainable AI business applications (in practice, those were days 6 and 7 during the programme, but for readability, we will call them Day 1 and Day 2).

4.2. Study setup and data collection

To facilitate our research, we framed the two-day activities around development of business ideas. This approach compelled participants to consider digital technology within practical and societal contexts, moving beyond an abstract, detached perspective. Our primary interest was not in the viability or quality of the business ideas themselves, but rather in what these ideas reveal about the participants' underlying values regarding digital technology and its design. By doing so, we aimed to access the deeper values that guide adolescents' perceptions and designs of technology.

On **Day 1**, the AI ambassadors were first provided with basic knowledge about business idea development during a brief lecture. Then, they were asked to discuss in small (random) groups how AI/robotics/digital tech might help to foster social innovations; take notes of their discussions; and present a three-minute summary. When presenting, they received feedback from the researchers.

Next, the AI ambassadors began generating business ideas individually, using the 'Crazy 8s' technique, sketching individually AI solution ideas for one minute. Crazy 8 is a rapid sketching technique where participants generate eight ideas in eight minutes to promote creative thinking. They were encouraged to focus on making a positive impact without judging ideas as good or bad. Afterward, they joined their teams and grouped similar ideas, combining them to create three concepts, and then selecting the most original and

valuable one to develop further. They were advised to enhance this idea by incorporating aspects of other concepts.

The teams then filled a business model canvas for quickly outlining business ideas (see e.g. Kinnula et al. (2024) for use with young people). We modified the original canvas by adding elements of value co-creation thinking (Vargo and Lusch 2016) to guide the AI ambassadors thinking towards value (benefit) creation for all stakeholders, i.e. creating a win-win situation for all who would be somehow involved/linked with the business idea. After working with the canvas, the teams drafted a textual business plan. The template included sections on the societal/sustainability challenge, their solution, target customers, benefits, marketing strategies, partnerships, resources, revenue streams, and risk assessment. In both the canvas and the business plan, ambassadors were prompted to identify key stakeholders, assess their benefits from the solution, and highlight the solution's broader positive impacts on the environment and society.

On **Day 2**, the teams finalised their business plans and spent time planning and rehearsing their presentations. These presentations, or 'pitches', were then delivered to a group of researchers who gave the teams feedback about their business ideas. They then reflected on their experiences over the past few days, discussing key learnings and insights that could be valuable to share with others. On both days, the ambassadors filled in a feedback questionnaire related to the activities.

4.3. Data analysis

Our study is qualitative, including a variety of empirical data (see Table 2). The analysis proceeded in two phases: (1) analysing initial ideation-related data, and (2) analysing business development-related data (the final business ideas that were worked in teams). This was done to gain an understanding of how values developed

Table 2. Data analysed from different phases of the work.

Ideation phase			Business Development phase		
Analysed data	Amount of data	Lenses used for data analysis	Analysed data	Amount of data	Lenses used for data analysis
Recordings of small group discussions and their presentations	4	Understanding good	Team business model canvases (one missing)	3	Understanding good
Crazy 8, individually produced ideas	12 sheets with 28 separate ideas		Team business plan text documents	4	Cosmic harmony (UN SDGs)
Daily feedback questionnaire answers	12		Team slide sets for business pitches (one missing)	3	Ethical triad
			Video recordings of teams' business pitches	4	
			Daily feedback questionnaire answers	12	

and changed during the process. The total duration of the videos we analysed, which includes team discussions and pitches, and subsequent discussions involving researchers, is 153 min and 20 s.

In analysing the Ideation phase, we categorised how the ideas, problems identified, and solutions presented look when reflecting the five Zoroastrian values on understanding good (see Table 3). Cosmic harmony and holistic alignment were not used in the ideation analysis as they are not feasible to analyse how implementation will impact cosmic harmony when the idea is acontextual, demarcated from a wider context. The ideas can be implemented in multiple ways, and not all of them align well with the idea of cosmic

harmony (some ideated solutions can be used for both positive and negative, like for example the World monitoring system one of the ambassadors presented). In our view, cosmic harmony analysis requires more information about the implementation, which can be derived from business plans. For the ethical triad analysis, we need to see how participants communicate (good words) about their ideas (good thoughts) and aim to put them into practice (good deeds). Business plans and pitches serve this need.

When analysing the Business Development phase, i.e. the final business ideas, we did it in three parts: (1) We started with the same Zoroastrian-inspired values analysis as for the Ideation phase. (2) That analysis was supplemented and strengthened by looking at how the final business ideas reflect in the Zoroastrian struggle towards cosmic harmony. For that purpose, we utilised UN SDGs. (3) Finally, we also looked at the business from perspective of the Zoroastrian ethical triad of 'good thoughts, good words, good deeds'.

Five researchers in total participated in designing the workshops and analysing the workshop data for this paper. Two researchers led the work with the adolescents during the workshop days. One researcher did the initial analysis of the material, but at a later stage, all five researchers participated in the analysis and paper writing. One of the participating researchers has Iranian heritage and existing knowledge of Zoroastrianism.

4.4. The use of AI

ChatGPT, Claude, and Copilot were used in writing this article. In the initial phase, AI was used as a discussion partner for ideation and conceptualisation of our Zoroastrian lens. These discussions were then reflected against the literature we used to understand Zoroastrian philosophy. Here, AI was especially useful for us

Table 3. Analysis framework.

Analysis lens	Sensitising questions for analysis
Understanding good: Integrity	To what extent authenticity, honesty, and wholeness of both the idea itself but also the process how it was developed are present?
Understanding good: Cooperation	To what extent do idea and implementation represent social harmony, cooperation, and collective welfare? How do parts work together to fulfil a purpose?
Understanding good: Sapience	How do idea and implementation reflect deeper wisdom, critical thinking, discernment, and ability to apply knowledge in a judicious manner in different situations?
Understanding good: Wellbeing	Do idea and implementation encompass physical or psychological well-being, mental peace, and spiritual elevation? (well-being is concerned not only with individuals but also with communities and their prosperity).
Understanding good: Protection of the environment	Do the idea and implementation aim to conserve and protect the natural world? How well does the idea recognise the interconnectedness of human existence and the well-being of different ecosystems?
Cosmic harmony	Are implications of shared responsibility or aims to build harmony visible in the data? In the context of the current study, we consider the struggle towards cosmic harmony to be the 17 UN SDGs
Ethical triad	What is the intention of the idea? How is the idea communicated, and what is the potential outcomes from the implementation of the idea? To what extent are these aligned with one another?

non-native English speakers in finding right nuances for the Zoroastrian concepts. However, we wrote the text related to the Zoroastrian lens ourselves based on the literature; no AI-generated text is included in the article.

In addition to supporting conceptual development, these tools were employed in assisting the article writing process to enhance the clarity, precision, and conciseness of the manuscript's language, with rigorous human review of the final text.

5. Findings

5.1. Predominance of practical and intelligence-oriented ideas in the ideation phase

We analysed 37 separate business ideas. Three different sources of data were used in that: presentations of the small group discussions (5 ideas); papers from the Crazy 8 exercise (28 ideas, see Figure 2 for an example of a Crazy 8 ideation sheet); and answers to the daily feedback questionnaire (4 ideas). It should be noted, however, that this is a rough division of the sources, as some ideas overlapped. Especially, Crazy8 papers contained many similar ideas. Many of the ideas, e.g. AI to predict future crises, were wide-ranging and could have been categorised under multiple values. Noting this, our approach was only to categorise each idea in a single category at this stage. For example, AI predicting future crises was categorised under sapience: the system helps to make better decisions, but could easily be categorised also under values such as environment. Table 4 presents how different ideas were distributed among different values. In naming the ideas we tried to use participant language to the extent possible.

From this first phase categorisation of ideas, we can see how participants focused on very practical issues instead of ethical or societal challenges. Many showed interest in everyday matters such as tuning instruments, designing clothing, or organising things at a store. Nevertheless, we wanted to extend our reading beyond the strict moral and ethical domain. For example, *integrity* can also be understood to include correctness and reliability in action, which helps explain why such technically oriented ideas can still embody integrity through their pursuit of order and precision (e.g. tuning instruments).

Looking at the values, we see that sapience – reflecting intelligence, deep understanding, and wisdom – was the most common value. The second most common value in this phase was well-being. Healthcare is a quite common context where AI is applied. Our participants were relatively young, and we are not sure to what extent they were familiar with such applications but considering how prevalent well-being and healthcare have been since the COVID-19 pandemic, this is quite natural value to be present in ideation. The third most common was cooperation, with ideas for improving communication between people. The remaining two values, integrity and protection of the environment, were equally present.

5.2. From ideas to values: developing ethically and cosmically aligned business concepts

For this second analysis phase, we deepened our analysis lens and adopted into our lens the struggle towards cosmic harmony and the Zoroastrian ethical triad. Next, we first present the final four business ideas developed in

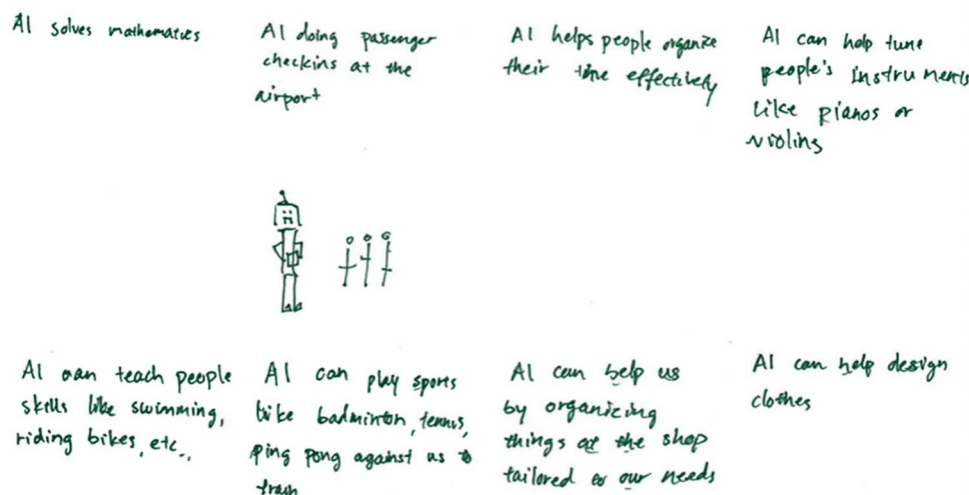


Figure 2. Ideas shown in one of the Crazy 8 papers.

Table 4. Business ideas categorised after the individual brainstorming.

Understanding good	Business ideas
Integrity (4 ideas)	AI to monitor what direction the world is going (politics, nature) AI to do passenger checks AI to improve hygiene and sanitation in food industry. AI to tune instruments
Cooperation (6 ideas)	AI to connect with people and participate in events. AI to recognise and translate languages live. AI to help nurses in all languages AI to connect to doctors who are not busy with patients. AI to optimise the location of services in a city Voice models to help mute people
Sapience (15 ideas)	AI to help organise time AI to help to calculate driving routes to save time, resources, and distance. AI to help with schoolwork by answering questions and helps to connect with other students and teacher AI to solve mathematics problems AI to organise things at a shop AI for time management. AI to predict future crises AI to optimise public transportation routes AI to help in choosing houses AI to train new skills AI to help in designing clothes AI as a law assistant AI for calculating the number of people in the city AI in literature and art AI to help understanding technology models
Wellbeing (8 ideas)	AI to play sports against humans AI to educate/employ unemployed people. AI to help in hospitals AI to chat with old people AI to act as therapist AI to monitor city traffic and help emergency response teams to move faster. AI to do gene synthesis for cancer research AI in mental health care
Protection of the environment (4 ideas)	AI to simulate the atmosphere to fight climate change Cleaning robots AI to improve recycling AI to simulate the impact of factories on the environment

the teams, and then our findings, divided by the dimensions of our analytical lens.

5.2.1 Four value-driven business concepts: from well-being apps to sustainable AI

The first business idea was *Vitalis*, an application that brings all aspects of well-being together and helps a user to become a better version of themselves (see Figure 3). The grand challenge the business idea solved was the confusion due to the complexity and mixed information one can find online and in different applications, i.e. providing *clear guidance for wellbeing in a world of mixed information*. This application works as a ‘one-stop-shop’ in providing guidance on daily

routines, including diet, exercise, and recovery. The way this application differs from its competitors is that the underlying AI utilises six different data when making suggestions: body type, lifestyle, age, background culture, training experience, and goals.

The second business idea, *SortAI*, aimed to improve trash handling at landfill sites and recycling centres (see Figure 4). The challenge the team recognised was the *inefficient waste management and resource utilisation* processes.

Their solution is an AI-based sorting chute where different types of trash are separated and put into their own containers. By doing so, *SortAI* allow more efficient use of resources due to recycling. In addition, the solution helps to save time and labour, making waste management more efficient.

Careo was the third business idea developed into a plan (see Figure 5). The team recognised that many patients do not get basic treatment due the challenge of *inadequate healthcare staff in hospitals and nursing homes*.

To tackle this challenge, the team envisioned a robot that would be an extension of the healthcare system that monitors and ensures health. The robot does not substitute nurses or doctors who still handle more complex tasks. Instead, the robot monitors and evaluates the status of patients, and alerts medical personnel if needed.

The last business idea was *Adminix* (see Figure 6). Their grand challenge was to tackle the *overburdened healthcare system*.

Their solution to the problem was a hospital management system that improves organisational capabilities at all levels. Utilising the system reduces the time healthcare workers spend on management tasks, making hospitals more efficient and enabling better patient service. The system works based on hospital records and patient data to make diagnoses, plans of action for patients, and equipment reservations, allowing workers to focus on more meaningful tasks in patient care.

Next, we explore the values-based underpinnings of these four business ideas, starting with the five values related to understanding good.

5.2.2 Values in Practice: How Each Business Concept Embodies ‘Understanding Good’

Vitalis prioritises *wellbeing* within the value framework, emphasising mental and physical health for a happier life. Its holistic approach aligns with *integrity* by not overstating wellbeing’s benefits and emphasising individual responsibility. As one of the participants noted, ‘And also some of the information is so contradicting. Like one says do this but then the other one says no do this’, and ‘So we have to be very realistic of what so people

- low threshold

location, analyse surroundings and available aspects

1. WHAT	2. TO WHOM	3. WHY	4. HOW	5. ASSESS
1.1 The challenge - not being able to adjust physical well-being according to own goals, not knowing how to - 16yrs and up - people who don't know how to integrate a healthier lifestyle/fitness goals - too much information available, contradicting, not sustainable or realistic	2.1 Customers - can divide based on goal, age, medical history, and current lifestyle, working out in what env. - sports watches linked to the application to the phone - app main technology used on phone	3.1 Benefits - improves physical wellbeing - simplifies the over-sharing of information online - people are more in touch with their health, may prevent need to go to hospital - easy to use, aimed at most people	4.1 Partnerships - medical professionals - sports watch companies - sport places 4.2 Key resources - trained AI - medical database - software - money, financing - knowledgeable people abt fitness and culture (food)	5.1 Risks [By your group] - AI analyses wrong - testing the data retrieval process before launching 5.2 To Improve [By your peers]
1.3 Your solution - an application that personalizes strategies for own physical wellbeing depending on personal goals - integrating medical and family background or e.g. diseases - applications that provide information similar to this - trainers - dieticians		3.2 Marketing - people turn to SM. for tips, so integrating our ads there can increase usage - Instagram, facebook, TV, newspapers	4.3 Revenue stream - all money invested in the beginning while developing AI	

Figure 3. Business Canvas of Vitalis.

don't expect too much from our programs'. These reflections show clear concern for honesty and realism in the advice, which we interpret as an expression of *integrity* understood as truthfulness, correctness, and consistency of action. They also illustrate *sapience*, as the team critically evaluates existing information and seeks to guide users through more informed and individually appropriate recommendations. While it lacks *cooperation* among users, the AI-human interaction is central. The application doesn't directly involve the environment,

yet informed users can leverage it to reduce dietary carbon footprint (*protection of the environment*).

SortAI, in contrast, emphasises values different from Vitalis. Primarily focused on improving waste management and recycling, it's closely tied to environmental concerns. Collaboration is key here, involving human employees to enhance recycling efficiency, highlighting the importance of *cooperation*. The team's reasoning summarises their motivation, 'But humans are lazy and there's additionally too much

The Challenge

Manual sorting of trash at landfill sites and recycling centres is inefficient & inaccurate leading it to be ecologically unsustainable.

- Dumping sites fill up quickly
- Unsustainable
- People are forced to work in trash sorting

An increasing amount of trash requires excessive amounts of workers and the workload leads to possible inaccuracies and slowness during the sorting process.

Currently the issue is "addressed" with manual labor which is the main issue we're trying to fix.

Search

Figure 4. The Challenge slide from the SortAI pitch presentation slide set.

Business Model

- We are great at business with years of experience.
- We have many ways of generating income:
 - Products
 - Development
 - Data Collection
 - Monthly subscriptions
- We can help you to develop AI technology.
- We are a publicly listed company and we could offer you:
 - A piece of our company.
- We offer what we make for what you give.
- We can decrease your fees.



Figure 5. Business Model of Careo as presented during their pitch.

trash. ... what is our solution to this? It is AI-based sorting machine. It sorts the trash really well. This reflection reveals pragmatic ethical stance, the recognition of human limitations, and the desire to design technology that supports collective responsibility for environmental care. In this sense, cooperation extends beyond teamwork to include human-machine collaboration aimed at shared ecological benefits. *Integrity* visible in the pursuit of accuracy and reliability in waste management, ensuring that the system fulfils its purpose truthfully. This ties into *sapience*, where informed decision-making is essential for efficient waste recognition. Ultimately, the system's success results in a cleaner environment, benefiting humans and other living beings' *wellbeing*.

Robot of **Careo** focuses on *wellbeing* by enhancing healthcare by offering augmented services and prompt patient care without replacing existing healthcare staff, emphasising *cooperation*. As one participant clarified their explicit intention to support rather than replace human professionals, 'Oh yeah, no, it doesn't like actually do the healing or anything. It just watches ... [the] robots would give them the knowledge and instructions'. This distinction reflects the moral awareness of technology's supportive rather than substitutive role in human-centred care, highlighting respect towards professional expertise and collaboration between humans and machines. The robot's ability to analyse and monitor patient status showcases intelligent decision-making, aligning with *sapience*, as it applies intelligent

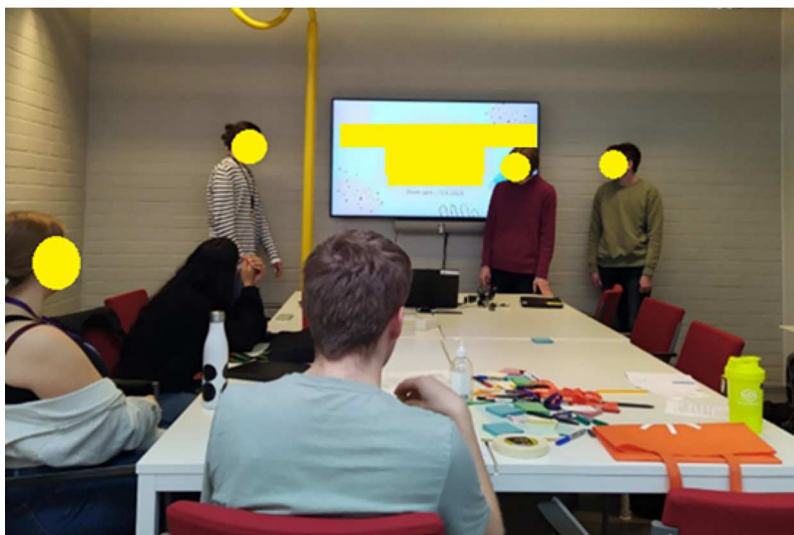


Figure 6. Adminix giving their presentation.

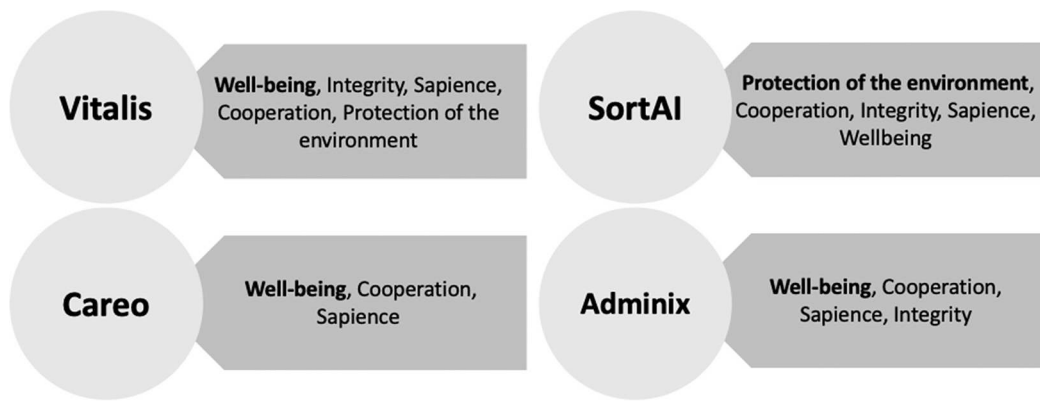


Figure 7. Summary of the dominant values in the business ideas.

decision-making to ensure effective and timely care. While the system's environmental impact isn't explicitly outlined, its timely care approach could indirectly benefit resource efficiency. However, environmental focus isn't a primary feature of the system.

The hospital management system, provided by **Adminix**, shares values with Vitalis and Careo robot, with a primary focus on *wellbeing*. Its goal is to enhance patient care and ease the workload for healthcare staff by fostering *cooperation* and handling administrative tasks. This aim was reflected in the team's discussions, '*So maybe it should be in nursing homes to help because there's a lack of nurses. So it would automate some processes there*'. This reasoning reveals an ethical awareness of workforce shortages and a desire to design technology that supports, rather than replaces, human care givers. By doing so, it aligns with cooperation as shared responsibility between humans and machines. The system's use of hospital records and patient data for decision-making demonstrate *sapience*, as it applies informed judgement to improve operational efficiency. *Integrity* is also central given the reliance on AI and handling sensitive nature of personal health data. While the system's implementation may indirectly influence the environment, it's not a primary concern within its design. As summary, we present the dominant values of each groups' business idea in [Figure 7](#).

5.2.2 Aligning with cosmic harmony: linking business ideas to global sustainability goals

Three business ideas were related to healthcare and well-being – Vitalis, Careo, and Adminix – making the business ideas well aligned with the Zoroastrian idea of cosmic order. SDG 3, 'Good Health and Well-being', aims to ensure healthy lives for all by emphasising everyone's right to access healthcare and medicines. The goal seeks universal health coverage and promotes research and development in health-related fields,

striving for a world where every individual can live a healthy life. While all three business ideas directly address SDG 3, their differences become visible when considering their secondary implications. Vitalis, in addition to being directly related to SDG 3, can also be seen as support for SDG 4, 'Quality Education', which emphasises lifelong learning opportunities for all through formal and informal learning. The sustainable lifestyle and diet choices that the application promotes relate to SDG 12, 'Responsible Consumption and Production'. Meanwhile, Careo and its robot relates to SDG 9, 'Industry, Innovation, and Infrastructure'. Robots can have a big impact on the medical infrastructure of societies. SDG 9 is also present in the hospital management system by Adminix. In addition, depending on how Careo robot and Adminix hospital management system are implemented, they can also be considered part of SDG11, 'Sustainable Cities and Communities' by providing solutions to the negative health impacts of air pollution and other health hazards of slums.

The remaining business idea SortAI focused on improving waste management. Environmental and sustainability issues are present in multiple SDGs, but the most relevant is SDG 12, 'Responsible Consumption and Production', that ensures sustainable consumption and production patterns. It emphasises the efficient use of resources, reducing waste through recycling and reuse, and promoting environmentally conscious behaviours and decisions in production and consumption to minimise the environmental impact and ensure long-term sustainability. In addition to the SDG 12, SortAI relates to multiple different SDGs. By providing novel solutions to waste management infrastructure, it is part of SDG 9. The system helps to reduce natural resource use and thus contributes to SDG 12. By improving waste handling, the system reduces the human impact on different ecosystems, thus contributes

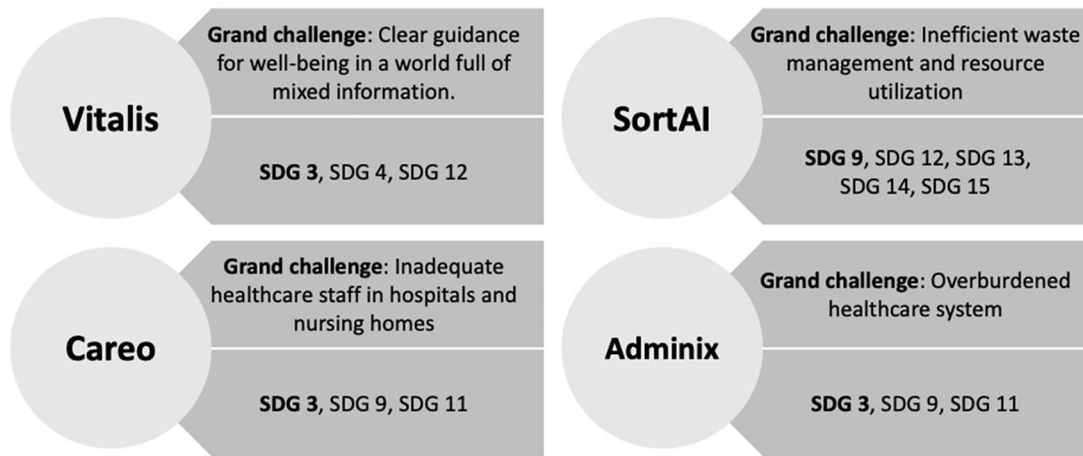


Figure 8. Business ideas and corresponding UN SDGs for cosmic harmony.

towards SDG 14, ‘Life Below Water’, and SDG 15, ‘Life on Land’. Finally, improved waste management can reduce greenhouse gas emissions such as methane from landfills, thus supporting SDG 13, ‘Climate Action’.

Taken together, the four business plans presented covered a total of eight different SDGs. By doing so, the ideas support the Zoroastrian cosmic struggle to create environmentally, morally, and socially harmonious world for everyone. As a summary, [Figure 8](#) presents the grand challenges the business ideas aimed to solve with their corresponding UN SDGs.

5.2.3 Living the ethical triad: good thoughts, good words, and good deeds in practice

In the case of Vitalis, the good thoughts come from a design that desires to improve one’s overall well-being. The application aims to reduce confusion and misinformation rampant on the Internet by providing a reliable source that everyone can turn to. The application’s message is also positive self-improvement, thus contributing to good words. It aims to replace a one-size-fits-all mentality with one that emphasises the individual’s unique needs through body type, lifestyle, age, culture, experience, and personal goals. This individually tailored experience enables users to become holistically better versions of themselves. The aim of the application is to turn these thoughts and positive messages into acts towards improving personal well-being. The healthier lifestyle that the application promotes will also benefit the society at large.

Regarding SortAI, good thoughts manifest through environmental consciousness and sustainable waste management driving the business idea. By emphasising the inefficiencies of existing solutions and aiming to rectify those inefficiencies, it demonstrates a forward-

looking mindset. This environmental progressiveness is also shown in their messaging and promotion of recycling and efficient resource utilisation. The practical good deeds come from implementing the suggested AI system to reduce waste and protect the environment. These are tangible steps towards a more sustainable and harmonious society.

Good thoughts are present in how Careo is looking for solutions to healthcare shortages. By doing so, they are looking for ways to ensure the well-being of the patients. This positive message is translated into good words when claiming that every patient is important and should not be left unmonitored. When put into practice, robots will result in tangible benefits for healthcare workers and patients alike. The solution has the potential to reduce errors in treatments and provide timely medical interventions, thus even saving patient lives.

Finally, Adminix shows good thoughts by designing a solution to inefficiencies of the healthcare system. This is motivated by the pressure and stress that the medical personnel endure. The core message they present is to improve organisational capabilities so that medical professionals can focus on the more important task of caregiving. As the solution provides streamlining administrative tasks, the system has the ability to decrease mistakes made in patient care, ensure timely care, and improve overall hospital efficiency.

Overall, all business ideas align well with the Zoroastrian ethical principle of ‘good thought, good words, and good deeds’. They recognise important challenges in existing ways of doing and propose solutions that benefit the wider society. Emphasis on well-being, healthcare, sustainability, and personalised care resonate well with the Zoroastrian ideals of truth, righteousness, and greater good. Summary of each groups’ holistic alignment is present in [Figure 9](#).

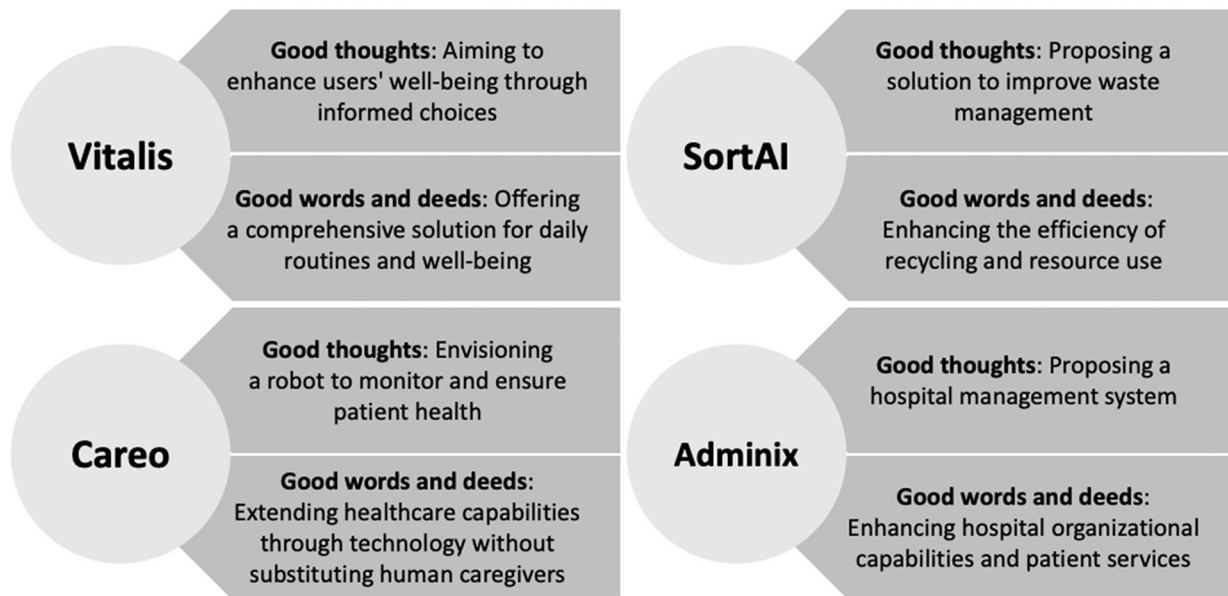


Figure 9. Business ideas and their holistic alignment.

6. Discussion

This paper examines how to foster responsible technology innovation among adolescents through the development of business ideas that utilise intelligent technologies, with the hope that they grow into responsible design protagonists who are able to create more sustainable solutions for the challenges of the future. We employed the Zoroastrian-inspired ethical lens, supplemented by the United Nations SDGs, to represent the struggle towards cosmic harmony, which is a notion characteristic of the Zoroastrian ethical philosophy. With our analysis, we show that certain Zoroastrian values were visible in the adolescents' business ideas. Next, we discuss how this understanding can inform strategies to foster responsible digital technology innovation in future design protagonists, proposing key elements for responsible digital technology design and innovation education for adolescents, holding the potential to enhance their critical and reflective thinking regarding future technologies.

6.1. Foregrounding the business plan revealed the embedded values

In our values analysis, we observed *correlations between highlighted values and the task at hand*. We also noted the importance of integrating innovation skills with technical knowledge. As progressed from ideation to business planning, their values shifted noticeably. Initially, AI was viewed mostly as a technological tool, but during business planning, there was a marked

emphasis on broader values, particularly well-being. The focus shifted from technology to considerations of social and ethical impacts, evident in healthcare and waste management ideas. This transition highlighted the need for a deeper understanding of societal challenges and innovative solutions that went beyond basic technological applications. As their ideas evolved, teenagers recognised the importance of creating solutions with both technological advancement and societal impact. AI ceased to be just a practical tool; it became a means to solve complex issues for the community at large.

We also noticed that the *business idea itself influences what values become foregrounded*. All the Zoroastrian-inspired values were present in the analysed business ideas, but the centrality of each value depended on the business idea. In all the cases, one of the values comes up as the primary one. This primary value was directly connected with the problem the system was trying to solve. For example, well-being is the primary value driving Careo robot. The rest of the values play a secondary role as an indirect effect or guiding principle for the system development. To continue with the Careo robot example, the environment can be indirectly affected, meanwhile sapience and integrity act as guiding principles. If we were analysing, for example, a business idea related to insurance, integrity or sapience might be the primary value of the system, and wellbeing acting guiding principle or just an indirect effect. Thus, the task description needs to be carefully considered beforehand if there is a wish for the participants to focus on certain values in their work.

The shift in values during our study and the inclusion of business perspectives carry important implications. *Equipping teenagers with both innovation skills and technical knowledge* empowers them to generate innovative ideas and build viable businesses capable of addressing complex societal challenges – a crucial skill in today's world. Complex problem-solving is an important skill in the twenty-first century. There is a need to delve deeper into the complex problems and uncertainties in the world to understand how humans tackle global and urgent challenges (Dörner and Funke 2017). Secondly, the focus on values in business ideas stresses the importance of *responsible AI innovation*. Encouraging teenagers to think about the wider implications of their tech creations fosters an early sense of responsibility, which is seen as essential with AI increasingly influencing society (Durall Gazulla et al. 2024). The ethics of AI is a global concern, engaging policymakers and researchers alike. Initiatives like the AI Governance Alliance by the World Economic Forum aim to promote responsible AI development, stressing the significance of ethical considerations. A lack of ethical awareness and unclear principles are noted as prominent obstacles when it comes to incorporating ethics into AI (Khan and Zaber 2022). Thirdly, value co-creation stresses *collaborative innovation*. Encouraging young individuals to collaborate with diverse stakeholders and adjust ideas based on real-world feedback promotes a more inclusive innovation approach. Our study illustrates how a business-oriented approach shapes values, facilitating a deeper comprehension of complex issues and fostering collaboration with multiple stakeholders, in line with Kinnula, Durall, and Haukipuro (2022) who propose learning systems thinking through business ideating.

See Table 5 for the key takeaways derived from our values analysis. Note that even though we used in our analysis the Zoroastrian value framework, we expect any value framework having a similar effect. I.e. the values embedded in a task description naturally guide teenagers' thinking – and also learning – towards that direction.

6.2. Chosen context and tools guide thinking

To shape responsible innovation thinking among adolescents, we focused on developing sustainable AI business plans (see e.g. Hartikainen et al. 2021; Kinnula, Durall, and Haukipuro 2022 for use of business ideating in children's design education context). Using an adapted Business Model Canvas aimed at creating value for all stakeholders guided their approach. This combination aligns with the societal demand for

Table 5. Key takeaways derived from our values analysis.

Aspect	Our recommendation for working with adolescents
Foregrounded values are linked with the given task	The researcher/practitioner should carefully consider beforehand what values adolescents should focus on in their work, and draft the task description accordingly; if the task guides towards finding solutions e.g. for sustainability, the values in the work naturally gravitate towards that direction. Being transparent with the used values framework, making it clearly visible to the adolescents, helps scaffold their critical and reflective thinking skills.
The business idea itself influences what values become foregrounded	Discussions and guidance around developing the business idea are central in guiding adolescents' thinking towards directions that serve best the given task. E.g. if the participants ideate a business linked with hospitals, the well-being of patients can become the central value, which might narrow the participants' thinking too much.
Linking innovation/entrepreneurial skills and knowledge alongside technical knowledge	Having only technical knowledge can leave ethical thinking superficial, therefore learning innovation/entrepreneurial thinking is recommended. We propose scaffolding a wider understanding of the complex intertwinement of technological innovations and business realities, in the societal context. It guides connecting the technology-focused ideas with a wider context, helping the shift towards exploring more nuanced and complex issues and values related to technology.
Responsible technological innovations	The task should encourage considerations of the broader impact of technological creations to help in taking responsibility when developing (technology-based) innovations.
The collaborative and networked nature of innovations	The task should (1) make visible the number of stakeholders/actors involved in both developing innovations as well as in using and maintaining them and encourage considerations of co-creating value (benefit) to all stakeholders, not only for some stakeholders, i.e. a win-win situation for everybody; and (2) encourage adolescents to engage with diverse stakeholders and adapt their ideas based on real-world needs and feedback, fostering a more inclusive innovation process.

sustainable AI solutions, offering a platform to develop ethical considerations related to technology in business contexts. Given the integration of AI in daily life, educating children about its ethical and societal impacts becomes crucial. There is a lot of previous work focusing on ethical AI education through school curricula,

projects, or other methods, such as game-based learning (Ali, Kumar, and Breazeal 2023), to teach children the societal impacts of AI systems. We propose this *contextual framework* as fruitful for ethical discussions.

The Business Model Canvas, a vital *tool in business planning*, facilitated linking tech-centred ideas to a broader context (cf. Kinnula et al. 2024 about learning systems thinking when business ideating). As the adolescents progressed from ideation to planning, their descriptions grew more comprehensive, reflecting collaboration with diverse stakeholders. Moving from ideation to planning demanded detailed task descriptions, signalling a commitment to creating value not just for themselves but also for end-users, customers, and society at large. This collaborative problem-solving approach extended benefits beyond their target customers to encompass healthcare providers, environmentalists, and society, aligning with the concept of value co-creation (Vargo and Lusch 2016) involving multiple stakeholders in the innovation process.

As for choosing a business context (compared to a pure technology context) and using the Canvas to guide thinking, we argue that it is essential to study intentionality and how plans evolve from an idea to a well-developed business concept (Krueger and Carsrud 1993). Incorporating a business perspective (compared to a single technology perspective), particularly through the Business Model Canvas, brought a multidimensional perspective into our analysis and enriched our insights. Firstly, the Business Model Canvas encouraged a holistic understanding of teenagers' ideas. While a technology perspective might primarily focus on the technical aspects of AI-driven innovations, the canvas enabled an examination of the entire business ecosystem surrounding these ideas. Secondly, the canvas bridged the gap between abstract technology concepts and practical applications, urging teenagers to evaluate their ideas' feasibility as real businesses. This shift *from conceptualising to practical implementation* added a realistic dimension to their analyses. Thirdly, it facilitated the integration of ethical considerations within a business framework. This required contemplating not only technical feasibility but also the ethical impact on customers, partners, and society – a crucial aspect in the AI era. Moreover, the canvas highlighted the interconnectedness of various business components, revealing potential synergies within the models and enabling a more nuanced analysis. From an educational standpoint, it underscored the importance of teaching not just technological innovation but also entrepreneurship and business fundamentals to young individuals. For a summary, refer to Table 6 outlining the key insights related to context and tools in innovation education.

Table 6. Key takeaways for choosing the context and tools.

Aspect	Our recommendation for working with adolescents
Contextual framework for business ideation	The contextual framework should be chosen to focus the participants' thinking towards current societal needs (e.g. creating sustainable AI solutions) and current and complex challenges, as they provide a good ground for developing ethical thinking in relation to technology use in society. We propose deriving those from UN SDGs, applying them in local and/or global situations, and linking them to the participants' lifeworld to increase relevancy and interest.
Adoption of business/innovation development tools	Proper tools used as scaffolds can improve understanding of complex issues and bring forward more nuanced perspectives. Here, we used a business model canvas, but depending on the context, the scaffold can differ. Tools for the strategic development of business can be useful here, too.
Practical implementation and realism	Business Model Canvas and other canvases can be used to translate abstract technology concepts into real-world business applications, fostering practicality and realism in entrepreneurial education. By doing so, canvases also bring forth the ethical dimensions of technological development, emphasising the necessity of aligning ethics with business practices in the age of AI.

6.3. What conscious use of leveraging Zoroastrian values can provide for innovation education for design protagonists

Based on our findings, we recommend leveraging Zoroastrian values (integrity, cooperation, sapience, well-being, and protection of the environment) in innovation education to guide adolescents' involvement with AI and future technologies, as the Zoroastrian values framework fosters human agency, societal harmony, and collective responsibility over societal and environmental implications of decisions that humans make (Boyce n.d.; Cantera 2015; Kreyenbroek n.d.). We think this value basis is interesting to apply in innovation education particularly with the notion of the design protagonist (Iversen, Smith, and Dindler 2017) in mind (Kinnula, Durall, and Haukipuro 2022), as the notion highlights a reflective stance towards digital technology design and use in society (Iversen, Smith, and Dindler 2017). In the following, we link the Zoroastrian values also more widely with the understanding of the skills and needs of entrepreneurs and business in more general, using AI as an example of a future technology.

Incorporating the value of 'integrity' into children's AI-related technological thinking serves as a platform

for their truthful understanding. Integrity enables children's accountability by underscoring their active roles and responsibilities such as promoting their thinking to respect for diversity, empathy, and compassion when it comes to the development and utilisation of AI technology. By considering the value of '*cooperation*', it is possible to elevate children's understanding of collaboration. Prioritising collaborative learning could provide opportunities for broadening children's horizons, wherein considering a global perspective by engaging in the inclusive design of AI and diversity could be considered an essential part of their collaborative learning. For entrepreneurs, cooperation is fundamental in building successful teams. '*Sapience*' value encourages critical thinking, problem-solving, and evidence-based decision-making, which are known to be important for entrepreneurs (Hsieh, Nickerson, and Zenger 2007) and promotes ethical AI design. Moreover, we recommend providing children with opportunities for effective role-taking, such as the AI ambassador role we gave the adolescents in our study. That role led to their engagement in effective communications with their peers as well as to their continuous learning. '*Well-being*' is a value that can be closely associated with the human-centred design of AI. In the context of AI, the initial thoughts revolve around mental and physical health considerations. We recommend nurturing the perspective of adolescents to also encompass a world-centred approach to wellbeing, including the aspects of environmental responsibility and societal contribution to wellbeing in both the development and utilisation of AI. From an entrepreneurial perspective, it is good to keep in mind that value-creation behaviour is always linked with well-being (Gong, Choi, and Murdy 2016; Grönroos and Voima 2013). Therefore, while taking care of the well-being of others provides entrepreneurial opportunities, as we have seen from our study, well-being as a value-guiding behaviour is fundamentally entwined with entrepreneurship. '*Protection of the environment*' could bring valuable insights into eco-friendly AI design thinking by considering, but not limited to, sustainable AI development, green innovations, green transportation, and waste management. Thus, the protection of the environment provides good opportunities for so-called 'ecological entrepreneurship' (Gast, Gundolf, and Cesinger 2017). Such an approach to entrepreneurship can help in providing competitive advantage (Kuo, Fang, and LePage 2022), which is a central question any innovator needs to consider.

In light of the '*cosmic harmony*' principle inherent in Zoroastrian thought, we advocate for its integration into the entrepreneurial mindset of children engaging with

AI. From the perspective of cosmic harmony, we are not just passive observers of societal forces, but rather, we have an important agency and role to play. Each individual is seen as an active architect in the shaping of society, with the power to influence and drive positive change. The perspective emphasises the interconnection and balance between all elements of existence, encouraging a holistic approach to AI development (Chiu 2021; Ng et al. 2022). In fostering an understanding of cosmic harmony, children can learn the importance of creating AI solutions that are harmonious not only within their immediate environment but also in the larger societal and natural world. This approach underscores the active role individuals play in maintaining societal and environmental equilibrium, mirroring the entrepreneurial spirit of innovation balanced with responsibility. From an entrepreneurial perspective, cosmic harmony aligns with the strategic thinking necessary for sustainable, long-term business success (Lozano 2018). It encourages young entrepreneurs to consider the broader implications of their innovations, fostering solutions that are sustainable, ethical, and beneficial on a societal level. This alignment with cosmic harmony can lead to innovations that not only serve immediate needs but also contribute positively to the future, ensuring longevity and relevance in a rapidly evolving world. Such a perspective is invaluable in entrepreneurial endeavours, where foresight, adaptability, and a deep understanding of the interconnectedness of various systems are key to sustainable success. Therefore, by embedding the essence of cosmic harmony in AI education, we enable children to develop a mindset that appreciates and strives for balance, interconnectivity, and the well-being of all. This holistic view is crucial not only for personal and professional development but also for nurturing responsible and visionary future entrepreneurs who can navigate the complexities of technology, society, and the environment with wisdom and foresight (Chiu 2021).

The principle of '*good thoughts, good words, good deeds*', central to Zoroastrianism, offers a profound framework for innovation education, particularly in the realm of AI and business. This triad encourages a mindset where ethical considerations, positive communication, and impactful actions are paramount (Gini 1997; Schwartz 2013). Fostering '*good thoughts*' guides children to think creatively yet responsibly, ensuring their ideas contribute positively to society without causing harm. This mindset of ethical innovation prioritises societal welfare and environmental consideration from the ideation phase. '*Good words*' stress clear, truthful, and constructive communication, vital for young innovators to express their ideas

Table 7. Key takeaways related to the use of the Zoroastrian values framework, with the context of sustainable development and use of AI as an example application of the lens.

Zoroastrian values	Our recommendation for working with adolescents
Integrity	In AI thinking, integrity is vital for promoting truthful understanding and accountability. It involves fostering respect for diversity, empathy, and compassion in AI development to overcome biases in AI systems, ensuring inclusive and ethical development. Educators should emphasise data quality by addressing factors like diversity, biases, and data security to underscore integrity in AI education.
Cooperation	In AI contexts, prioritising cooperation fosters collaborative learning, broadens global perspectives, and emphasises inclusive AI design by considering diverse viewpoints. It underscores the significance of team building and stakeholder relationships for successful AI endeavours. Encouraging interactive tasks involving diverse groups can effectively cultivate cooperation among learners.
Sapience	Sapience entails promoting ethical AI use by encouraging critical thinking, problem-solving, and evidence-based decision-making among learners. Roles like AI ambassadors can facilitate hands-on engagement and ongoing learning in the AI field. Assignments involving ethical dilemmas and diverse perspectives effectively nurture sapience in learners.
Well-being	To integrate well-being into AI, adopt human-centred and environmentally responsible design, emphasising societal contributions. Connect well-being with value-driven entrepreneurship to ensure AI benefits individuals and the community. Engaging adolescents in activities emphasising mutual responsibility and societal well-being effectively highlights our collective contribution to a better society.
Protection of the environment	Advocating environmental protection in AI involves promoting eco-friendly design, sustainable development, and responsible waste management. Encouraging green innovations not only benefits the environment but also offers competitive advantages in branding and stakeholder relations. Incorporating environmental protection in learning includes exercises that unveil the environmental impact of daily activities and seek solutions to reduce this impact.
Cosmic harmony	Teaching cosmic harmony emphasises balance, interconnectivity, and societal-environmental equilibrium. It encourages strategic thinking for sustainable success and advocates holistic development across personal, professional, and entrepreneurial aspects. Engaging young individuals in analysing various ecosystems fosters a comprehensive understanding of their significance.
Ethical triad	Incorporating the holistic alignment of good thoughts, words, and deeds into AI and business education means instilling an ethical innovation mindset from ideation. It stresses clear, truthful communication and ethical persuasion in development and interaction. It also emphasises practical, beneficial, user-friendly solutions grounded in ethical principles. Alongside business plan exercises, case study analyses serve as effective tools for learning this holistic alignment.

effectively and collaborate transparently. Teaching this aspect encourages persuasive and ethical presentation, fostering an environment of openness and trust. Lastly, ‘good deeds’ in innovation education translate to practical application, ensuring developed technologies are not just innovative but also beneficial, accessible, and sustainable. This approach instils a sense of responsibility, emphasising that their creations should positively impact the world.

Refer to [Table 4](#) for key insights regarding the conscious application of Zoroastrian values in innovation education.

Compared with established value-oriented approaches in design such as Value-Sensitive Design (Friedman et al. 2013), Worth-Centered Design (Cockton 2006), and Values-Led Participatory Design (Leong and Iversen 2015), our Zoroastrian-inspired framework contributes a distinctive cosmological and ethical dimension. While the aforementioned frameworks provide structured methods for incorporating stakeholder values, experiential worth, and participatory ethics into design processes, they remain largely procedural and anthropocentric, focusing on human actors and negotiated values within bounded design contexts.

The Zoroastrian lens, by contrast, situates design intentions within a holistic moral and cosmological order through the notions of *cosmic harmony* and the ethical triad of *good thoughts, good words, and good deeds*. This perspective connects the personal, societal, and environmental dimensions of design in a unified

moral framework. Rather than replacing existing models, it complements them by offering an ontological grounding that extends value-sensitive design from the social domain to include ecological and moral coherence thus linking responsible technological innovation with the pursuit of what is conceived as a universal good within the Zoroastrian tradition.

This ontological grounding has methodological implications because it shapes how relevance is assigned during design. In practical terms, values help explain why certain design concerns ranging from environmental impact and moral responsibility to long-term societal effects are treated as relevant or sometimes even non-negotiable within a design process. For example, in a Value Sensitive Design – informed workshop, values identified through stakeholder discussions (e.g. efficiency or user autonomy) can be further examined through the Zoroastrian lens to reflect on their alignment with longer-term ecological responsibilities and moral consequences beyond immediate user needs. Different value frameworks therefore guide designers’ attention toward different evaluative horizons, even when using the same procedural methodology.

6.4. Limitations to the study

The participants in the study were selected on a first-come, first-served basis. Not all of them were native to the city or country, and almost half of them were

studying in an English-speaking International Baccalaureate Diploma Programme for their senior high school diploma. They all spoke fluently the local language, and a host of other languages with some participants being fluent in 3–4 languages, while learning a couple more. Not all participants were interested in business, nor did all have a background in information technology, but they all mentioned an interest in learning more about AI through the AI Ambassadors programme and having a paid summer job position. While not representative of the general young population in the city, they formed a unique group. However, the study's relevance lies in its findings related to the working process, rather than the characteristics of the participants.

Our study intentionally did not gather or analyse demographic data such as the participants' gender. This is because our participants were self-selected, meaning they chose to join the study on their own. In such a self-selection scenario, using demographic data like gender for analysis is not very meaningful or representative, as it does not necessarily reflect a broader population. Furthermore, our study's focus was on group behaviour rather than individuals. While it is acknowledged that gender and sex differences can influence values (Atari, Lai, and Dehghani 2020; Connolly et al. 2020; Schwartz and Rubel 2005), these aspects were not pertinent to the goals of our current study. Future research, however, could investigate how these demographic differences impact the generation of business ideas.

While the Zoroastrian ethical triad and the notion of cosmic harmony provided a meaningful analytical lens for interpreting our data, it also represents a specific religious-philosophical worldview. Drawing on a religious framework in a largely practical and secular field such as design may invite scepticism. We fully recognise that researchers or educators from other spiritual or secular traditions might approach value-based design through different ethical constructs, such as liberal humanism, Buddhism, or secular virtue ethics. Recognising this plurality is essential: the Zoroastrian framework should be understood as one possible pathway for exploring moral and cosmological coherence in design, rather than a prescriptive model. Our framework is not intended to be universal, but rather an invitation to dialogue between traditions, encouraging a pluralistic rather than singular worldview.

7. Conclusion

We chose the Zoroastrian lens complemented with UN SDGs to understand how we can foster adolescents to

grow into such 'innovators for the digital futures' who are capable of responsible technology and business thinking. The Zoroastrian-inspired lens helped us to uncover deeper ethical dimensions within the design ideas generated by teenagers. It provided a structured framework for analysing the values embedded in these ideas, going beyond surface-level considerations. Moreover, by applying the Zoroastrian-inspired lens across different stages of the study, it became evident how the values transformed and evolved from initial ideation to the development of business plans. This evolution provided us with valuable insights into how adolescents' ethical consciousness developed as they progressed from abstract concepts to more practical business proposals.

We contribute by answering the call for finding ways of educating business-oriented design protagonists with responsible mindsets (Hartikainen et al. 2023a, 2023b). We propose conscious and transparent instrumentalisation of values in the innovation process, using the Zoroastrian values framework as one illustrative and productive lens within this broader effort. From our perspective, it offers one articulation of a mindset that is increasingly relevant in a world shaped by complex and often invisible digital technologies with significant ethical implications for their design and use. Ethical and reflective stance towards digital technology is in the core of the design protagonist notion, and we think the Zoroastrian values framework can support researchers and practitioners working with young people by helping them gain deeper understanding of how to solve the 'big issues' (Bødker and Kyng 2018) we are facing. We call for future research to explore Zoroastrian values alongside other philosophical, cultural, and ethical frameworks, examining how diverse moral ontologies may jointly inform young people's involvement with AI, or any other technology.

Note

1. United Nations Sustainable Development Goals (SDGs). <https://sdgs.un.org/goals> (accessed November 10, 2023).

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Ethics approval

According to the ethical review board of the authors' university, this study did not require formal review, as none of the criteria mandating such review were met. Informed consent was obtained from all participants and their legal guardians prior to participation.

Consent

Informed consent was obtained from all participants and their legal guardians prior to participation in the study. Participants were informed of the purpose of the study, their right to withdraw at any time, and how their data would be handled.

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References

- Ali, S., V. Kumar, and C. Breazeal. 2023. *AI Audit: A Card Game to Reflect on Everyday AI Systems*. <https://doi.org/10.48550/ARXIV.2305.17910>.
- Ames, M. G., D. K. Rosner, and I. Erickson. 2015. "Worship, Faith, and Evangelism: Religion as an Ideological Lens for Engineering Worlds." *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing*: 69–81. <https://doi.org/10.1145/2675133.2675282>.
- Atari, M., M. H. C. Lai, and M. Dehghani. 2020. "Sex Differences in Moral Judgements across 67 Countries." *Proceedings of the Royal Society B: Biological Sciences* 287 (1937): 20201201. <https://doi.org/10.1098/rspb.2020.1201>.
- Bødker, S., and M. Kyng. 2018. "Participatory Design That Matters – Facing the Big Issues." *ACM Transactions on Computer-Human Interaction* 25 (1): 4:1–4–4:131. <https://doi.org/10.1145/3152421>.
- Boyce, M. 1997. "The Origins of Zoroastrian Philosophy." In *Companion Encyclopedia of Asian Philosophy*, edited by B. Carr and I. Mahalingam, 4–20. London and New York: Routledge.
- Cantera, A. 2015. "Ethics." In *The Wiley Blackwell Companion to Zoroastrianism*, edited by M. Stausberg, Y. S. Vevaina, and A. Tessmann, 313–332. Chichester, UK: Wiley. <https://doi.org/10.1002/9781118785539.ch19>.
- Chiu, T. K. F. 2021. "A Holistic Approach to the Design of Artificial Intelligence (AI) Education for K-12 Schools." *TechTrends* 65 (5): 796–807. <https://doi.org/10.1007/s11528-021-00637-1>.
- Cockton, G. 2006. "Designing Worth Is Worth Designing." *Proceedings of the 4th Nordic Conference on Human-Computer Interaction: Changing Roles*: 165–174. <https://doi.org/10.1145/1182475.1182493>.
- Connolly, F., Goossen, M., & Hjerem, M. (2020). Does Gender Equality Cause Gender Differences in Values? Reassessing the Gender-Equality-Personality Paradox. *Sex Roles*, 83 (1–2): 101–113. <https://doi.org/10.1007/s11199-019-01097-x>.
- Conrad, Michael, and M. A. Rahimi. 1984. *Computers and the Future of Human Creativity*. National computer Conference and Exposition. <https://doi.org/10.1145/1499310.1499370>.
- Dhalla, M. N. 1938. *History of Zoroastrianism*. New York: Oxford University Press.
- Dörner, D., and J. Funke. 2017. "Complex Problem Solving: What It Is and What It Is Not." *Frontiers in Psychology* 8: 1153. <https://doi.org/10.3389/fpsyg.2017.01153>.
- Durall Gazulla, E., N. Hirvonen, S. Sharma, H. Hartikainen, V. Jylhä, N. Iivari, and M. Kinnula. 2024. "Youth Perspectives on Technology Ethics: Analysis of Teens' Ethical Reflections on AI in Learning Activities." *Behaviour & Information Technology* 44 (5): 888–911. <https://doi.org/10.1080/0144929X.2024.2350666>.
- Flanagan, M., D. C. Howe, and H. Nissenbaum. 2008. "Embodying Values in Technology: Theory and Practice." In *Information Technology and Moral Philosophy*, edited by J. van den Hoven and J. Weckert, 322–353. Cambridge, UK: Cambridge University Press. <https://doi.org/10.1017/CBO9780511498725.017>.
- Foltz, R. C. 2002. "Iran's Water Crisis: Cultural, Political, and Ethical Dimensions." *Journal of Agricultural and Environmental Ethics* 15 (4): 357–380. <https://doi.org/10.1023/A:1021268621490>.
- Friedman, B., P. H. Kahn, A. Borning, and A. Hultgren. 2013. "Value Sensitive Design and Information Systems." In *Early Engagement and New Technologies: Opening up the Laboratory*, edited by N. Doorn, D. Schuurbiers, I. van de Poel, and M. E. Gorman, 55–95. Dordrecht: Springer Netherlands. https://doi.org/10.1007/978-94-007-7844-3_4.
- Gast, J., K. Gundolf, and B. Cesinger. 2017. "Doing Business in a Green way: A Systematic Review of the Ecological Sustainability Entrepreneurship Literature and Future Research Directions." *Journal of Cleaner Production* 147: 44–56. <https://doi.org/10.1016/j.jclepro.2017.01.065>.

- Geser, G., Eva-Maria Hollauf, V. Hornung-Prähauser, S. Schön, and F. Vloet. 2019. "Makerspaces as Social Innovation and Entrepreneurship Learning Environments: The DOIT Learning Program." *Discourse and Communication for Sustainable Education* 10 (2): 60–71. <https://doi.org/10.2478/dcse-2019-0018>.
- Gini, A. 1997. "Moral Leadership: An Overview." *Journal of Business Ethics* 16 (3): 323–330. <https://doi.org/10.1023/A:1017959915472>.
- Gong, T., J. N. Choi, and S. Murdy. 2016. "Does Customer Value Creation Behavior Drive Customer Well-Being?" *Social Behavior and Personality: An International Journal* 44 (1): 59–75. <https://doi.org/10.2224/sbp.2016.44.1.59>.
- Grönroos, C., and P. Voima. 2013. "Critical Service Logic: Making Sense of Value Creation and Co-creation." *Journal of the Academy of Marketing Science* 41 (2): 133–150. <https://doi.org/10.1007/s11747-012-0308-3>.
- Haddington, P., N. Hirvonen, S. Hosio, M. Kinnula, J. Malmberg, S. Seyfi, J. Simonen, S. Ahola, M. Cortés Orduna, H. Enwald, L. Haukipuro, M. Heikkinen, J. Hermes, S. Huikari, N. Iivari, S. Järvelä, O. Kanste, L. Kokkola, S. Kunnari, and K. Zabolotna. 2021. *GenZ White Paper: Strengthening Human Competences in the Emerging Digital era*. Oulu, Finland: University of Oulu. <http://urn.fi/urn:isbn:9789526231471>.
- Hagendorff, T. 2020. "The Ethics of AI Ethics: An Evaluation of Guidelines." *Minds and Machines* 30 (1): 99–120. <https://doi.org/10.1007/s11023-020-09517-8>.
- Hannafey, F. T. 2003. "Entrepreneurship and Ethics: A Literature Review." *Journal of Business Ethics* 46 (2): 99–110. <https://doi.org/10.1023/A:1025054220365>.
- Harris, J. D., H. J. Sapienza, and N. E. Bowie. 2009. "Ethics and Entrepreneurship." *Journal of Business Venturing* 24 (5): 407–418. <https://doi.org/10.1016/j.jbusvent.2009.06.001>.
- Hartikainen, H., N. Iivari, and M. Kinnula. 2019. "Children's Design Recommendations for Online Safety Education." *International Journal of Child-Computer Interaction* 22: 100146. <https://doi.org/10.1016/j.ijcci.2019.100146>.
- Hartikainen, H., L. Ventä-Olkkonen, N. Iivari, S. Sharma, E. Lehto, T. Molin-Juustila, and J. Holappa. 2023a. "We Learned We Can Do Something to Reduce Bullying: Children Designing Anti-bullying Mobile Apps to Empower Their Peers." *Frontiers in Education* 8: 1112835. <https://doi.org/10.3389/feduc.2023.1112835>.
- Hartikainen, H., L. Ventä-Olkkonen, M. Kinnula, and N. Iivari. 2021. Entrepreneurship Education Meets FabLab: Lessons Learned with Teenagers. *FabLearn Europe / MakeEd 2021 – An International Conference on Computing, Design and Making in Education*. <https://doi.org/10.1145/3466725.3466727>.
- Hartikainen, H., L. Ventä-Olkkonen, M. Kinnula, and N. Iivari. 2023b. "We Were Proud of Our Idea': How Teens and Teachers Gained Value in an Entrepreneurship and Making Project." *International Journal of Child-Computer Interaction* 35: 100552. <https://doi.org/10.1016/j.ijcci.2022.100552>.
- Hollauf, E.-M., V. Hornung-Prähauser, D. Podmetina, E. Unterfrauner, and G. Geser. 2020. "Making Social Innovators – Novel Innovation Education for Youth in Makerspaces." In *ISPIM Conference Proceedings*, 1–15. ABI/INFORM Collection.
- Hsieh, C., J. A. Nickerson, and T. R. Zenger. 2007. "Opportunity Discovery, Problem Solving and a Theory of the Entrepreneurial Firm." *Journal of Management Studies* 44 (7): 1255–1277. <https://doi.org/10.1111/j.1467-6486.2007.00725.x>.
- Huang, T.-F., and P.-H. Lin. 2021. "On the Zen Connotation in Product Design." In *Cross-Cultural Design. Experience and Product Design across Cultures*, edited by P.-L. P. Rau12771, 300–310. Cham, Switzerland: Springer International Publishing. https://doi.org/10.1007/978-3-030-77074-7_24.
- Ibtasam, S. 2021. "For God's sake! Considering Religious Beliefs in HCI Research: A Case of Islamic HCI." *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*: 1–8. doi:10.1145/3411763.3450383.
- Iivari, N., O. S. Iversen, R. C. Smith, M.-M. Schaper, L. Ventä-Olkkonen, H. Hartikainen, S. Sharma, et al. 2024. "Transformative Agency – the Next Step towards Children's Computational Empowerment." *Proceedings of the 23rd Annual ACM Interaction Design and Children Conference*: 322–337. <https://doi.org/10.1145/3628516.3655806>.
- Iivari, N., M. Kinnula, and L. Kuure. 2015. "With Best Intentions: A Foucauldian Examination on Children's Genuine Participation in ICT Design." *Information Technology & People* 28 (2): 246–280. <https://doi.org/10.1108/ITP-12-2013-0223>.
- Iivari, N., T. Molin-Juustila, and M. Kinnula. 2016. *The Future Digital Innovators: Empowering the Young Generation with Digital Fabrication and Making*.
- Iivari, N., L. Ventä-Olkkonen, H. Hartikainen, S. Sharma, E. Lehto, J. Holappa, and T. Molin-Juustila. 2023. "Computational Empowerment of Children: Design Research on Empowering and Impactful Designs by Children." *International Journal of Child-Computer Interaction* 37: 100600. <https://doi.org/10.1016/j.ijcci.2023.100600>.
- Isomursu, M., M. Ervasti, M. Kinnula, and P. Isomursu. 2011. "Understanding Human Values in Adopting new Technology – a Case Study and Methodological Discussion." *International Journal of Human-Computer Studies* 69 (4): 183–200. <https://doi.org/10.1016/j.ijhcs.2010.12.001>.
- Iversen, O. S., K. Halskov, and T. W. Leong. 2010. "Rekindling Values in Participatory Design." *Proceedings of the 11th Biennial Participatory Design Conference*: 91–100. <https://doi.org/10.1145/1900441.1900455>.
- Iversen, O. S., and R. C. Smith. 2012. "Scandinavian Participatory Design: Dialogic Curation with Teenagers." *Proceedings of the 11th International Conference on Interaction Design and Children*: 106–115. <https://doi.org/10.1145/2307096.2307109>.
- Iversen, O. S., R. C. Smith, and C. Dindler. 2017. "Child as Protagonist: Expanding the Role of Children in Participatory Design." *Proceedings of the 2017 Conference on Interaction Design and Children*: 27–37. <https://doi.org/10.1145/3078072.3079725>.
- Jobin, A., M. Ienca, and E. Vayena. 2019. "The Global Landscape of AI Ethics Guidelines." *Nature Machine Intelligence* 1 (9): 389–399. <https://doi.org/10.1038/s42256-019-0088-2>.

- Kayal, A., M. B. van Riemsdijk, M. A. Neerincx, and W.-P. Brinkman. 2018. "Socially Adaptive Electronic Partners for Improved Support of Children's Values: An Empirical Study with a Location-Sharing Mobile app." *International Journal of Child-Computer Interaction* 18: 79–89. <https://doi.org/10.1016/j.ijcci.2018.09.001>.
- Khan, S., and M. Zaber. 2022. "Assessing the State of E-Governance Responsiveness at Local Level: A Case Study of 1st Level Administrative Units of India, Nepal and Bhutan." *15th International Conference on Theory and Practice of Electronic Governance*: 381–390. <https://doi.org/10.1145/3560107.3560166>.
- Kheirandish, S., M. Funk, S. Wensveen, M. Verkerk, and M. Rauterberg. 2020. "HuValue: A Tool to Support Design Students in considering Human Values in Their Design." *International Journal of Technology and Design Education* 30 (5): 1015–1041. <https://doi.org/10.1007/s10798-019-09527-3>.
- Kinnula, M., E. Durall, and L. Haukipuro. 2022. "Imagining Better Futures for Everybody – Sustainable Entrepreneurship Education for Future Design Protagonists." *6th FabLearn Europe / MakeEd Conference 2022*: 1–8. <https://doi.org/10.1145/3535227.3535229>.
- Kinnula, M., E. Durall, N. Hirvonen, J. Malmberg, and L. Haukipuro. 2024. "Nurturing Systems Thinking among Young People by Developing Business Ideas on Sustainable AI." *International Journal of Child-Computer Interaction* 40: 100656. <https://doi.org/10.1016/j.ijcci.2024.100656>.
- Kinnula, M., E. Durall Gazulla, N. Hirvonen, J. Malmberg, and L. Haukipuro. 2024. "Nurturing Systems Thinking among Young People by Developing Business Ideas on Sustainable AI." *International Journal of Child-Computer Interaction* 40: 100656. <https://doi.org/10.1016/j.ijcci.2024.100656>.
- Kinnula, M., and N. Iivari. 2021. "Manifesto for Children's Genuine Participation in Digital Technology Design and Making." *International Journal of Child-Computer Interaction* 28: 100244. <https://doi.org/10.1016/j.ijcci.2020.100244>.
- Kinnula, M., N. Iivari, M. Isomursu, and H. Kinnula. 2018. "Socializers, Achievers or Both? Value-Based Roles of Children in Technology Design Projects." *International Journal of Child-Computer Interaction* 17: 39–49. <https://doi.org/10.1016/j.ijcci.2018.04.004>.
- Kreyenbroek, P. G. 1997. "Morals and Society in Zoroastrian Philosophy." In *Companion Encyclopedia of Asian Philosophy*, edited by B. Carr and I. Mahalingam, 42–59. London and New York: Routledge.
- Krueger, N. F., and A. L. Carsrud. 1993. "Entrepreneurial Intentions: Applying the Theory of Planned Behaviour." *Entrepreneurship & Regional Development* 5 (4): 315–330. <https://doi.org/10.1080/08985629300000020>.
- Kuo, F.-I., W.-T. Fang, and B. A. LePage. 2022. "Proactive Environmental Strategies in the Hotel Industry: Eco-Innovation, Green Competitive Advantage, and Green Core Competence." *Journal of Sustainable Tourism* 30 (6): 1240–1261. <https://doi.org/10.1080/09669582.2021.1931254>.
- Kynigos, C., and N. Yiannoutsou. 2018. "Children Challenging the Design of Half-Baked Games: Expressing Values through the Process of Game Modding." *International Journal of Child-Computer Interaction* 17: 16–27. <https://doi.org/10.1016/j.ijcci.2018.04.001>.
- Laato, S., S. Rauti, and J. Hamari. 2023. "Resemblance of Religion and Pervasive Games: A Study among Church Employees and Gamers." *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*: 1–15. <https://doi.org/10.1145/3544548.3581056>.
- Leong, T. W., and O. S. Iversen. 2015. "Values-led Participatory Design as a Pursuit of Meaningful Alternatives." *Proceedings of the Annual Meeting of the Australian Special Interest Group for Computer Human Interaction*: 314–323. <https://doi.org/10.1145/2838739.2838784>.
- Lindfors, E., and A. Hilmola. 2016. "Innovation Learning in Comprehensive Education?" *International Journal of Technology Design Education* 26 (3): 373–389. <https://doi.org/10.1007/s10798-015-9311-6>.
- Lozano, R. 2018. "Sustainable Business Models: Providing a More Holistic Perspective." *Business Strategy and the Environment* 27 (8): 1159–1166. <https://doi.org/10.1002/bse.2059>.
- Martin, K. E., and R. E. Freeman. 2004. "The Separation of Technology and Ethics in Business Ethics." *Journal of Business Ethics* 53 (4): 353–364. <https://doi.org/10.1023/B:BUSI.0000043492.42150.b6>.
- Miller, J. K., B. Friedman, G. Jancke, and B. Gill. 2007. "Value Tensions in Design: The Value Sensitive Design, Development, and Appropriation of a Corporation's Groupware System." *Proceedings of the 2007 International ACM Conference on Supporting Group Work*: 281–290. <https://doi.org/10.1145/1316624.1316668>.
- Naamneh, S., N. Atamni, B. Madi, D. Vasyutinsky Shapira, I. R. Rabaev, J. El-Sana, and S. Boardman. 2023. "Classifying the Scripts of Aramaic Incantation Bowls." *Proceedings of the 7th International Workshop on Historical Document Imaging and Processing*: 55–60. <https://doi.org/10.1145/3604951.3605510>.
- Nagda, H., S. P. Singhal, M. J. Amiri, and B. T. Loo. 2024. "Rashnu: Data-Dependent Order-Fairness." *Proceedings of the VLDB Endowment* 17 (9): 2335–2348. <https://doi.org/10.14778/3665844.3665861>.
- Neha, K., V. Agrawal, V. Kumar, T. Mohan, A. Chopra, A. B. Buduru, R. Sharma, and P. Kumaraguru. 2022. A Tale of Two Sides: Study of Protesters and Counter-Protesters on #CitizenshipAmendmentAct Campaign on Twitter. *14th ACM Web Science Conference 2022*, 279–289. <https://doi.org/10.1145/3501247.3531584>.
- Ng, M. Y., S. Kapur, K. D. Blizinsky, and T. Hernandez-Boussard. 2022. "The AI Life Cycle: A Holistic Approach to creating Ethical AI for Health Decisions." *Nature Medicine* 28 (11): 2247–2249. <https://doi.org/10.1038/s41591-022-01993-y>.
- Nigosian, S. A. 1993. *Zoroastrian Faith: Tradition and Modern Research*. Montreal and Kingston: McGill-Queen's University Press. <https://www.jstor.org/stable/j.ctt814hf>.
- Nouwen, M., M. Van Mechelen, and B. Zaman. 2015. "A Value Sensitive Design Approach to Parental Software for Young Children." *Proceedings of the 14th International Conference on Interaction Design and Children*: 363–366. <https://doi.org/10.1145/2771839.2771917>.
- Paço, A. do, and M. J. Palinhas. 2011. "Teaching Entrepreneurship to Children: A Case Study." *Journal of*

- Vocational Education & Training* 63 (4): 593–608. <https://doi.org/10.1080/13636820.2011.609317>.
- Park, W., and H. Kwon. 2024. “Implementing Artificial Intelligence Education for Middle School Technology Education in Republic of Korea.” *International Journal of Technology and Design Education* 34 (1): 109–135. <https://doi.org/10.1007/s10798-023-09812-2>.
- Read, J. C., D. Fitton, G. Sim, and M. Horton. 2016. “How Ideas Make It through to Designs: Process and Practice.” *Proceedings of the 9th Nordic Conference on Human-Computer Interaction*: 1–10. <https://doi.org/10.1145/2971485.2971560>.
- Rifat, M. R., F. A. Peer, H. Rabaan, N. J. Mim, M. Mustafa, K. Toyama, R. B. Markum, et al. 2022. “Integrating Religion, Faith, and Spirituality in HCI.” *CHI Conference on Human Factors in Computing Systems Extended Abstracts*: 1–6. <https://doi.org/10.1145/3491101.3503705>.
- Rifat, M. R., T. Toriq, and S. I. Ahmed. 2020. “Religion and Sustainability: Lessons of Sustainable Computing from Islamic Religious Communities.” *Proceedings of the ACM on Human-Computer Interaction* 4 (CSCW2): 1–32. <https://doi.org/10.1145/3415199>.
- Rutkowski, T. L., H. Hartikainen, K. E. Richards, and P. J. Wisniewski. 2021. “Family Communication: Examining the Differing Perceptions of Parents and Teens regarding Online Safety Communication.” *Proceedings of the ACM on Human-Computer Interaction* 5 (CSCW2): 373:1–373:23. <https://doi.org/10.1145/3479517>.
- Schwartz, M. S. 2013. “Developing and Sustaining an Ethical Corporate Culture: The Core Elements.” *Business Horizons* 56 (1): 39–50. <https://doi.org/10.1016/j.bushor.2012.09.002>.
- Schwartz, S. H. 1992. “Universals in the Content and Structure of Values: Theoretical Advances and Empirical Tests in 20 Countries.” In *Advances in Experimental Social Psychology*, edited by M. P. Zanna, 1–65. San Diego, CA: Academic Press. [https://doi.org/10.1016/S0065-2601\(08\)60281-6](https://doi.org/10.1016/S0065-2601(08)60281-6).
- Schwartz, S. H. 1994. “Are There Universal Aspects in the Structure and Contents of Human Values?” *Journal of Social Issues* 50 (4): 19–45. <https://doi.org/10.1111/j.1540-4560.1994.tb01196.x>.
- Schwartz, S. H., and T. Rubel. 2005. “Sex Differences in Value Priorities: Cross-Cultural and Multimethod Studies.” *Journal of Personality and Social Psychology* 89 (6): 1010–1028. <https://doi.org/10.1037/0022-3514.89.6.1010>.
- Shrestha, P., P. Delir Haghighi, G. Sharp, J. Xie, and R. McNaney. 2024. “One Size (Doesn’t) Fit All: Exploring Design Considerations for Digital Body Dissatisfaction Interventions with Underrepresented Populations.” *Proceedings of the 2024 ACM Designing Interactive Systems Conference*: 3534–3550. <https://doi.org/10.1145/3643834.3661539>.
- Stausberg, M. 2008. “On the State and Prospects of the Study of Zoroastrianism.” *Numen* 55 (5): 561–600. <https://doi.org/10.1163/156852708X310536>.
- Vaida, M.-F. 2019. “Collaborative Education Teams Development Using Alternative Methodologies.” *Proceedings of the 2019 11th International Conference on Education Technology and Computers*: 223–227. <https://doi.org/10.1145/3369255.3369280>.
- Vallaster, C., S. Kraus, J. M. Merigó Lindahl, and A. Nielsen. 2019. “Ethics and Entrepreneurship: A Bibliometric Study and Literature Review.” *Journal of Business Research* 99:226–237. <https://doi.org/10.1016/j.jbusres.2019.02.050>.
- Vargo, S. L., and R. F. Lusch. 2016. “Institutions and Axioms: An Extension and Update of Service-Dominant Logic.” *Journal of the Academy of Marketing Science* 44 (1): 5–23. <https://doi.org/10.1007/s11747-015-0456-3>.
- Ventä-Olkkonen, L., H. Hartikainen, B. Norouzi, N. Iivari, and M. Kinnula. 2019. “A Literature Review of the Practice of Educating Children about Technology Making.” In *Human-Computer Interaction – INTERACT 2019*, edited by D. Lamas, F. Loizides, L. Nacke, H. Petrie, M. Winckler, and P. Zaphiris, 418–441. Cham, Switzerland: Springer International Publishing.
- Ventä-Olkkonen, L., M. Kinnula, H. Hartikainen, and N. Iivari. 2021. “Embedded Assumptions in Design and Making Projects with Children.” *Proceedings of the 32nd Australian Conference on Human-Computer Interaction*: 178–188. <https://doi.org/10.1145/3441000.3441077>.
- Wilson, K. E., S. Vyakarnam, C. Volkmann, S. Mariotti, and D. Rabuzzi. 2009. “Educating the Next Wave of Entrepreneurs: Unlocking Entrepreneurial Capabilities to Meet the Global Challenges of the 21st Century (SSRN Scholarly Paper ID 1396704).” *Social Science Research Network*. <https://doi.org/10.2139/ssrn.1396704>.
- Winschiers-Theophilus, H., and N. J. Bidwell. 2013. “Toward an Afro-Centric Indigenous HCI Paradigm.” *International Journal of Human-Computer Interaction* 29 (4): 243–255. <https://doi.org/10.1080/10447318.2013.765763>.
- Wisniewski, P., A. K. Ghosh, H. Xu, M. B. Rosson, and J. M. Carroll. 2017. “Parental Control vs. Teen Self-regulation: Is There a Middle Ground for Mobile Online Safety?.” *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing*: 51–69. <https://doi.org/10.1145/2998181.2998352>.
- World Economic Forum. n.d. *New Vision for Education – Unlocking the Potential of Technology*. Retrieved March 15, 2021, from <http://widgets.weforum.org/nve-2015/>.
- Wyche, S. P., and R. E. Grinter. 2009. “Extraordinary Computing: Religion as a Lens for Reconsidering the Home.” *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*: 749–758. <https://doi.org/10.1145/1518701.1518817>.
- Yachkaschi, A., and S. Yachkaschi. 2012. “Nature Conservation and Religion.” *Forest Policy and Economics* 20:107–111. <https://doi.org/10.1016/j.forpol.2011.12.002>.